

# Foreword

Digital technology is all around us, giving us access to information, communication, and entertainment that would have seemed unimaginable to people 100 years ago. Computer science is the study of how this technology works, from the microchips at the heart of devices, to the code that controls them. Studying computer science gives young people the tools to understand today's technology and puts them in a position to create the machines, apps, websites, and services of the future. Rather than being restricted by what others create, computer science gives students the ability to turn their own ideas into reality.

The technology of the future will benefit from having a diverse array of people develop and shape it, and the creators of that technology will need to be able to work with others, communicate effectively, and also have some great ideas.

The skills developed in understanding computer science are useful even to those not intending to specialize in computing. Some of the main lessons of computer science, such as breaking complex problems down into sections and seeing patterns in them, are skills that are useful in any career. Understanding the core concepts of how computers work is fast becoming a necessity for many careers, even if they are not directly related to making technology.

As computers continue to play an increasing role in the modern world, we must also think about the way we use them. Social networking has revolutionized the way we communicate with each other, but it's worth being aware of the potential problems that we can face – from too much screentime, to social media bubbles, and even to being victims of cyberbullying. This book also tackles issues that the internet age has ushered in, from the digital divide, to net neutrality, and diversity and inclusion in the digital world.